

How many ways can you give out k if you have only one

Because I have only 1 coin I it means I have only one way to give out 0,1,2,3,4,5,6 ...11

Introducing 2 give me more ways because I already have previous v

Dynamic Programming Coin Change – Number of ways to Make Sum Coins[1, 2, 3]. Sum = 11.												
Coins/sum	0	1	2	3	4	5	6	7	8	9	10	11
1	1	1	1	1	1	1	1	1	1	1	1	1
2	1	1	2									
	1											

Coins/sum	0	1	2	3	4	5	6	7	8	9	10	11
1	1	1	1	1	1	1	1	1	1	1	1	1
2	1	1	2	2	3	3	4	4	5	5	6	6
	1											

Coins/sum	0	1	2	3	4	5	6	7	8	9	10	11
1	1	1	1	1	1	1	1	1	1	1	1	1
2	1	1	2	2	3	3	4	4	5	5	6	6
3	1	1	2	3	4	5	7	8	10	12	14	16

6-3 =3 number of ways for 3

The **distinct ways** coins can be dispersed for a value sum can be computed using the recursive formula.

if sum == 0:

1 (only one way)

else if sum > 0:

$\text{diffWays}(i, \text{sum}) = \text{diffWays}(i, \text{sum} - \text{coin}[i]) + \text{diffWays}(i - 1, \text{sum})$

where, $0 \leq i \leq m - 1$ and $\text{coins}[i] \leq \text{sum}$.

Dynamic Programming Coin Change – Number of ways to Make Sum Coins[1, 4, 5]. Sum = 12.													
Coins/sum	0	1	2	3	4	5	6	7	8	9	10	11	12
	1												
	1												
	1												

Coins/sum	0	1	2	3	4	5	6	7	8	9	10	11	12
1	1	1	1	1	1	1	1	1	1	1	1	1	1
	1												
	1												

Coins/sum	0	1	2	3	4	5	6	7	8	9	10	11	12
1	1	1	1	1	1	1	1	1	1	1	1	1	1
4	1	1	1	1	2	2	2	2	3	3	3	3	4
	1												

Coins/sum	0	1	2	3	4	5	6	7	8	9	10	11	12
1	1	1	1	1	1	1	1	1	1	1	1	1	1
4	1	1	1	1	2	2	2	2	3	3	3	3	4
5	1	1	1	1	2	3	3	3	4	5	6	6	7